Locating Zeros and Turning-Point Constraints

Factoring to locate zeros, reading multiplicity against degree, and bounding the number of turning points

Directions

- Factor completely before listing zeros — a zero you cannot see in the factored form is a zero you will miss.
- The multiplicities of the zeros add up to the degree, and a polynomial of degree n has at most $n - 1$ turning points.
- Give exact zeros. Leave fractions as fractions.

1. Find all real zeros of $p(x) = x^2 - 7x + 12$.

Answer: ________________

2. A polynomial has degree 6. What is the greatest possible number of turning points its graph can have?

Answer: ________________

3. Let $p(x) = (2x-1)^2(x+4)$.

(a) Find all real zeros of p.

Answer: ____________________

(b) State the multiplicity of each zero and whether the curve crosses or touches the axis there.

Answer: ____________________

4. Find all real zeros of $p(x) = x^3 - 3x^2 - 4x$.

Answer: ____________________

5. Let $p(x) = x^4 - 3x^2 + 1$.

(a) What is the greatest possible number of turning points of its graph?

Answer: ____________________

(b) State $\lim\limits_{x \to -\infty} p(x)$.

Answer: ____________________

6. Let $p(x) = x^3 + x^2 - 6x$.

(a) Factor p completely.

Answer: ____________________

(b) List all real zeros.

Answer: ____________________

7. Let $p(x) = (x+1)^2(x-3)^3$.

(a) What is the degree of p?

Answer: ____________________

(b) What is the greatest possible number of turning points of its graph?

Answer: ____________________

8. Find all real zeros of $p(x) = x^4 - 13x^2 + 36$.

Answer: ____________________

9. A polynomial of degree 5 with a positive leading coefficient has exactly three real zeros, each of multiplicity 1.

(a) What is the greatest possible number of turning points of its graph?

Answer: ____________________

(b) Explain why the graph must have at least two turning points.

Answer: ____________________

10. Let $p(x) = 2x^3 - 3x^2 - 11x + 6$. One of its zeros is $x = 3$.

(a) Divide out the factor $(x - 3)$ and factor p completely.

Answer: __________________

(b) List all real zeros.

Answer: __________________

11. Let $p(x) = x^5 - 4x^3$.

(a) Factor p, then list all real zeros.

Answer: __________________

(b) Give the multiplicity of each zero and say which ones the curve crosses.

Answer: __________________

12. A polynomial p has degree 4, a zero at $x = -2$ of multiplicity 2, simple zeros at $x = 1$ and $x = 3$, and $p(0) = 24$. So $p(x) = a(x+2)^2(x-1)(x-3)$.

(a) Use $p(0) = 24$ to find a.

Answer: ________________

(b) State the greatest possible number of turning points, then explain how many the graph actually has.

Answer: ________________